Bayesian Statistics III: Bayesian Linear Regression

ECON 8310: Business Forecasting — Lecture 12

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Lecture Outline

Bayesian Linear Regression

The same regression you already know — returning a distribution for every coefficient.

OLS vs. Bayesian Regression: What Changes?

You have fitted linear models since Lecture 2, and regularized them in Lecture 6. OLS solves for a single best coefficient vector:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

One number per predictor, with a standard error resting on large-sample theory. The Bayesian version replaces that with the update rule from Lecture 10:

$$p(\boldsymbol{\beta} \mid \mathbf{y}) \;\propto\; p(\mathbf{y} \mid \boldsymbol{\beta}) \cdot p(\boldsymbol{\beta})$$

> **Every coefficient comes back as a distribution.** Not an estimate plus an error bar computed from an asymptotic formula — a full posterior you can ask questions of directly.

And one thing you know reappears: **regularization is a prior**. A Normal prior on β is Ridge; a Laplace prior is LASSO. Lecture 6's penalty is, here, a statement of belief.

Bayesian Linear Regression in PyMC

Log units on log price, SNAP and events

```
with pm.Model() as model:
a      = pm.Normal("a", y.mean(), 1)
b_price = pm.Normal("b_price", 0, 1)     # weakly informative
b_snap = pm.Normal("b_snap", 0, 1)
b_event = pm.Normal("b_event", 0, 1)
sigma   = pm.Exponential("sigma", 1.0)
mu = a + b_price*lpc + b_snap*snap + b_event*event
pm.Normal("obs", mu=mu, sigma=sigma, observed=y)
idata = pm.sample(1000, tune=1500, chains=4)
```

Do not standardize log price — with both in logs `b_price` is an **elasticity**, percent change in demand per percent change in price. `lpc` is log price centered within series, which fixed ESS 184 $\rightarrow$ 3,097.

Reading the Posterior

Three questions you can ask a distribution that you cannot ask a point estimate.

Reading a Posterior Coefficient

Fitted on the weekly panel, controlling for series and trend:

Coefficient	Mean	94% HDI	Reading
`b_price`	-0.493	$[-0.596,\ -0.387]$	Near the elastic boundary
`b_trend`	0.110	$[\ 0.107,\ \ 0.113]$	About 11% growth per year
`b_snap`	0.036	$[\ 0.027,\ \ 0.046]$	About a 3.7% SNAP-week lift
`b_event`	-0.003	$[-0.011,\ \ 0.004]$	**Spans zero**

The **HDI** — highest density interval — is the shortest interval containing 94% of the posterior. Unlike a confidence interval it says what everyone wants it to say: there is a 94% probability the coefficient lies in there.

The last row is the useful one. `b_event`'s interval spans zero, so the data do not establish a direction. That is a more honest report than "not significant," because it also shows the effect is *small* — at most about half a percent either way.

Asking the Posterior a Direct Question

The posterior is a set of samples, so any question about β is answered by counting.

One line each

```
b = idata.posterior["b_price"].values.ravel()
(b < 0).mean()            # P(demand falls when price rises)
(b < -0.5).mean()         # P(elasticity beyond -0.5)
((b > -0.5) & (b < 0)).mean()  # P(inelastic but negative)
```

On our fit, $P(\text{elasticity} < 0) = \mathbf{1.00}$ — the direction is not in doubt. But $P(\text{elasticity} < -0.5) = \mathbf{0.446}$: the data cannot settle whether demand is elastic or inelastic.

Two findings, and a p-value conflates them. “Significant” reports only the direction. The question the business asks — *how big?* — gets the answer 46/54.

*Posterior mass inside a range of negligible values is a **ROPE** — region of practical equivalence.*

Causal Structure and DAGs

Which coefficients mean what you think they mean.

Directed Acyclic Graphs

A **DAG** is a picture of what you believe causes what: **nodes** are variables, **arrows** run from cause to effect, and **acyclic** means nothing causes itself.

It is drawn before the model, from domain knowledge, and its purpose is to tell you *which variables belong in the regression*.

Confounder	Causes both your predictor and your outcome. **Must** be controlled for, or its influence is absorbed into your coefficient.
Collider	Is caused by both. Controlling for it **creates** a spurious association that was not there.

That second row is why "add every control you have" is bad advice. Which variables to include is a question about causal structure, and no amount of model fit will answer it.

Whether a given DAG is right is a substantive question about your business, not a statistical one. This course uses DAGs to reason about what a coefficient means — estimating causal effects credibly is a separate subject.

Confounding, Measured

Our panel has thirty store-category series. Categories differ in *both* typical price and typical volume — so series identity is an arrow into price *and* into units. A textbook confounder.

Fit the same elasticity twice:

Model	**Elasticity**	**94% HDI**
Price, SNAP, events only	-2.805	$[-2.856,\ -2.756]$
$+$ series effects and trend	$\mathbf{-0.493}$	$[-0.596,\ -0.387]$

The uncontrolled estimate is nearly six times too large — it claims demand collapses when price rises. It is not noisier either: its interval is *half the width*. It is confidently, precisely wrong.

> A tight credible interval says the model is sure. It says nothing about whether the model is answering your question. Bayesian machinery does not protect you from omitting a confounder — only the DAG does.

Scenario Analysis with Posterior Samples

Because the posterior is samples, you can push a business scenario through it and get a **distribution of outcomes** rather than a single projected number.

What if we raise price 5%?

```
b = idata.posterior["b_price"].values.ravel()
pct_change = 100 * (np.exp(b * np.log(1.05)) - 1)
pct_change.mean(), np.percentile(pct_change, [3, 97])
```

With an elasticity near -0.493, a 5% price rise implies roughly a **2.4% fall in units**, 94% of the posterior between -2.9% and -1.9% — so you can report the pessimistic end too.

Then the decision follows. Revenue rises if the volume loss is smaller than the price gain: here $+2.5\%$ on average, and $P(\text{revenue rises}) = 1.00$ across every posterior sample. Note this holds *even though* the elasticity might be past -0.5.

Valid only to the extent the DAG is — this is a causal claim about intervening on price.

Course Synthesis and Next Steps

From classical time series to Bayesian hierarchical models: a complete forecasting toolkit.

Course Method Map

Method	Data	Uncertainty	Interpretable	Strengths
ETS / ARIMA (L01–02)	Univariate TS	CI	High	Fast, automatic
GAM / Prophet (L03)	TS + predictors	Limited	High	Nonlinear + interpretable
Decision Tree (L04)	Tabular	No	High	Rule-based explanation
Random Forest (L05)	Tabular	OOB CI	Medium	Robust, minimal tuning
XGBoost (L06)	Tabular	No	Low	Best tabular accuracy
FFN / 1D CNN (L07–08)	Tabular / TS	No	Low	Smooth functions, fast
LSTM / Transformer (L09)	Sequences	No	Very low	Long-range dependencies
Bayesian (L10–12)	Any	Full posterior	High	Uncertainty, small samples

No single method dominates. Choose based on: (1) data type and size, (2) need for uncertainty quantification, (3) need for interpretability, (4) time to deploy. In practice, combine: use XGBoost for the point forecast and a Bayesian model for uncertainty intervals.

Lecture 12: Key Takeaways

1. **Bayesian linear regression** places priors on coefficients and returns a full posterior distribution for each β_j. A Normal prior is equivalent to Ridge regularization; a Laplace prior is equivalent to LASSO.
2. Posterior coefficient plots give: posterior mean, HDI, and $P(\beta_j > 0)$ — richer information than a single p-value.
3. **DAGs** make the causal structure explicit, helping identify confounders (must include) and colliders (must exclude). Always draw the DAG before specifying a regression model.
4. **Scenario analysis** with posterior samples: plug any counterfactual $\mathbf{x}^*$ into posterior draws to get a full predictive distribution for that scenario.
5. **No single method is best for all forecasting problems.** Build intuition for when to use each tool in this course's toolkit, and combine methods (e.g., point forecast + Bayesian uncertainty) when the decision stakes are high.

References I